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Analysis of muscle activation in upper limb during a functional task: towards a practical use of EMG

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ABSTRACT

This bachelor thesis aims at establishing a reference pattern by conducting a systematic analysis of upper-limb muscle activation during a functional reach-and-grasp task using surface electromyography (sEMG). The objective was to characterize the typical activation patterns of key upper-limb muscles and assess their consistency within and between healthy individuals. sEMG signals were recorded from multiple muscles as participants performed repeated reach-to-grasp movements. Signals were temporally normalized to the movement cycle and amplitude-normalized to each muscle's maximum, then averaged to produce representative activation curves. Cross-correlation analysis quantified the similarity of these muscle activation profiles across repeated trials (intra-subject) and between different subjects (inter-subject).

The results demonstrate a highly consistent muscle activation pattern. Within individuals, activation timing and profiles were highly reproducible across repetitions. Across subjects, a common activation sequence emerged with only minor variations. Proximal muscles (e.g., triceps and front deltoid) showed particularly high repeatability, whereas distal muscles exhibited slightly greater variability but still adhered to an overall pattern. These findings indicate the existence of a normative muscle activation pattern for the reach-and-grasp task. Establishing this reference pattern has important clinical implications, as it can be used to benchmark patient performance, monitor rehabilitation progress, and detect atypical activation patterns in neuromuscular disorders.

RESUMEN

Este trabajo de fin de grado tiene como objetivo realizar un análisis sistemático de la activación muscular del miembro superior durante una tarea de alcance y agarre, utilizando electromiografía de superficie (EMG). El objetivo fue caracterizar los patrones típicos de activación y evaluar su consistencia tanto en un mismo individuo como entre individuos sanos. Se registró la actividad EMG de varios músculos mientras los sujetos repetían la tarea. Las señales se normalizaron en tiempo y amplitud antes de promediarlas en curvas de activación representativas. Se empleó la correlación cruzada para cuantificar la similitud de los perfiles de activación entre repeticiones del mismo sujeto y entre sujetos distintos.

Los resultados revelan un patrón de activación altamente consistente. Dentro de cada individuo, la secuencia de activación se repitió con gran fidelidad. Entre sujetos, surgió un patrón común con variaciones mínimas. Los músculos proximales mostraron mayor repetibilidad que los distales, aunque estos últimos también conservaron el patrón general. Estos hallazgos sugieren la existencia de un patrón normativo de activación muscular para la tarea de alcance y agarre. Dicho patrón de referencia podría tener aplicaciones clínicas, como base para comparar el desempeño de pacientes, monitorear la rehabilitación y detectar activaciones atípicas en trastornos neuromusculares.

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1 INTRODUCTION

1.1 Importance of reach and grasp movements in everyday life

The ability to perform Activities of Daily Living (ADLs) is fundamental for individuals to maintain an autonomous and independent life. Among these basic activities (eating, grooming, dressing, handling objects at home, etc.), the upper limb's function and especially the hand's ability to reach and grasp objects is essential. The functional task of reaching and grasping consists of moving the hand towards an object (reach phase) and exerting pressure on the object to pick it up (grasp phase). These two phases are normally accompanied by another two phases that complete or complement the movement, the first of these phases is moving the grasped object to a desired position (transport phase) and finally depositing said object someplace (release phase). This coordinated movement involves multiple articulations and muscular groups from the upper limb ranging from the proximal muscles of the shoulder to the intrinsic muscles of the hand. During the reaching and grasping phases we can mainly see a shoulder flexion accompanied by an extension of the elbow and wrist joints, preparing our arm's position to grasp the desired object; the sequence continues with grasping phase in which the fingers close around the desired object, activating both the finger flexors and extensors in order to hold it in a secure manner. After this first half of the movement, the grasped object is then deposited back in its initial position by inverting the order of the phases I just mentioned. The correct execution of this reach and grasp movement is essential for ADLs such as drinking from a cup, opening a door, or manipulating tools, therefore, alterations in this gesture or task can severely affect a person's functional autonomy [1]. To better understand the importance of reaching and grasping in upper limb motion analysis, we will now be looking at the golden standard on neuromuscular system evaluation [2].

1.2 Electromyographic analysis of the gait cycle: a consolidated reference

In the last few decades, gait cycle analysis has become the golden standard tool for neuromotor analysis in neurorehabilitation [3]. Gait is considered one of the most important ADLs and its study provides valuable information about a subject's global motor function. Particularly, surface electromyography (sEMG) during gait is routinely employed to detect alterations in muscle activation patterns of patients with neurological or orthopedic pathologies, and for evaluating the effectiveness of therapeutical or surgical interventions designed to improve gait. Measuring EMG activity along with other kinematic and kinetic analysis during gait allows us to better understand and identify compensations or muscular deficits that may not be evident through clinical observation alone [4], [5].

One of the strengths of gait analysis is the availability of normative data that serves as reference. In gait analysis laboratories and clinical biomechanics units, it is common to represent average muscle activation curves throughout the gait cycle for healthy patients, establishing normality bands (mean ± standard deviation). These curves provide an activation pattern against which we can compare the curves of an individual patient. This way, the clinician can interpret an EMG recording by asking, "How does this pattern compare to normal?"; the existence of reliable reference data allows them to identify atypical deviations in the patient's data and quantify the magnitude of the alteration. Thanks to this objective information, gait EMG analysis contributes to guiding clinical decision making, such as diagnosing abnormal muscle coordination, planning gait rehabilitation, or assessing changes after a certain treatment. In conclusion, gait analysis represents an exemplary model of how biomechanical instrumentation, and the use of normative patterns have enriched functional assessment in clinical settings [5].

1.3 The need to systematically analyze functional movements of the upper limb

Unlike gait, there is not yet a widely established equivalent for the functional analysis of the upper limb using EMG. This is relevant given that gestures such as reach and grasp have a comparative relevance to gait in ADLs and exhibit significant biomechanical complexity. Historically, upper limb function evaluation (especially after neurological lesions such as ictus or trauma) has been based on clinical scales and standardized functional tests – for example, the Fugi-Meyer scale, the Frenchy Arm Test, the Nine-Hole Peg Test or box and block – that assess the capacity and dexterity when performing certain tasks. These tools provide a global measure of function, sensitive enough to detect gross deficits, but they may not be sufficiently sensitive to subtle changes in muscle coordination or movement quality [6]. In other words, two patients could obtain the same functional score despite exhibiting very different muscle activation patterns to achieve the task.

In this context, it is justified to develop a systematic analysis of functional upper limb movements using EMG, analogous to gait analysis. Counting with objective instrumental registers will allow this study to dig deeper into the characterization of how the arm's muscles activate during complex movements, detecting specific alterations (no matter how small they may be) and quantify the effects of rehabilitative interventions with greater precision. Moreover, establishing standardized and non-biased EMG datasets eases the comparison between studies and laboratories, improving reproducibility and allowing for the development of benchmarking protocols. This, in turn, supports international collaborations by providing a common methodological framework upon which future clinical and technological research can be based. This will also benefit models that try to integrate the data this study will produce into the robotics field [7].

The reach and grasp gesture is an ideal candidate for this: it is representative of the functionality of the upper limb, reproducible in the laboratory and relevant to numerous daily tasks [8]. Just as a cycle is defined in gait (from heel contact to the next contact of the same heel) to average and compare patterns, a similar movement cycle can be defined for reach and grasp (for example, from the start of reaching to the completion of the grasping) and apply temporal and amplitude normalization techniques that will allow us to obtain a representative average pattern. Systematizing this procedure would provide normative curves for the upper limb, facilitating comparisons between healthy subjects and patients, as well as between different time points for the same patient as follow-up. To achieve this, we must ensure the maximum repeatability between repetitions and participants. To this end, cross-correlation analysis will provide a quantitative repeatability measurement.

1.4 Cross-correlation as a Tool for Comparing EMG Signals

In the dynamic analysis of muscle activity, the clinical interpretation of EMG signals is often based on qualitative assessments (e.g., visual observation of the moments of muscle activation) [9]. As an objective alternative, the cross-correlation technique allows us to quantitatively compare the shape and timing of two time-normalized EMG signals. Wren et al. (2006) [10] demonstrated that this method yields high correlation coefficients when comparing repeated trials from the same individual under similar conditions (in their gait study, average R ≥ 0.9 between different strides within the same session), and even when comparing trials from different sessions conducted on the same subject. These findings indicate that cross-correlation can robustly validate the consistency of intra-subject activation patterns.

In contrast, when comparing different subjects, this gait-based study observed much greater variability in the correlation coefficients (average R ranging from 0.4 to 0.81, with extreme values from 0 to 0.91), highlighting a limitation of this technique for direct inter-subject comparisons. In other words, the cross-correlation was found to be useful when trying to detect changes or notable differences in the activation pattern of a patient across a period of time (for example, before and after surgery), but it was found to be less suitable for comparing absolute patterns between different subjects. Even with these limitations, cross-correlation offers some clear advantages: it considers the full shape of the EMG signal and provides objective evaluations, reducing reliance on visual inspection or the clinician's subjective judgment [10]. Therefore, in the context of this project, cross-correlation is proposed as an objective and robust tool to validate the repeatability of EMG patterns within the same individual, providing greater rigor than simple visual comparison – though its results should be interpreted with caution when extended to comparisons between subjects.

1.5 Current state of EMG in reach and grasp tasks

In scientific literature, various exploratory studies have been conducted on muscle activation during functional upper limb tasks, including variations of the reach and grasp gesture [1], [7], [11], [12]. However, most of these studies have addressed very concrete aspects of the movement or have employed heterogeneous methodologies which hinders the extraction of a general pattern. Jarque-Bou et al. (2021), in a systematic review of forearm and hand EMG during ADLs, pointed out the lack of standardized protocols and consistent methodologies among studies, which hinders the comparison of results between different investigations and among different subjects [2]. In fact, many previous studies concentrate on specific muscles and/or movements (e.g., studying a single grip, or isolated finger movements) and are affected by a lack of complete functional tasks representativity [7], [11]. This lack of a unified approach means that, to date, a "typical" or normative activation pattern for the reach and grasp gesture has not been established, which could serve as a clinical or research reference.

Despite this, some important precedents can be cited that demonstrate the interest and potential reach and grasp EMG analysis has. For example, Bonnefoy et al. (2009) studied the influence different distances to the object had on the activation patterns of various muscles of the arm on several healthy patients during a reach and grasp task in a seated position [7]. During their experiment, they varied the distance that separated the patient from the object between three different configurations (20, 30 and 40 cm) and concluded, after registering EMG data for five muscles, that the level of muscle activation systematically increases as the distance to be reached increases [7]. Additionally, they proved that their measurement protocol presented high intra-operator repeatability (intraclass correlation coefficients, ICC, ranging from 0.78 to 0.99), which indicates it is possible to obtain reliable and consistent EMG activity measurements from this gesture in a controlled environment. The results Bonnefoy et al. produced contribute to identifying muscle activation levels in a simple reach and grasp task, knowledge considered essential for calculating muscle forces and integrating this data into biomechanical grasp models used in robotics and rehabilitation.

A different approach is provided by Liarokapis et al. (2013). Who applied machine learning algorithms to recognize reach and grasp gestures based on EMG signals. In their study, they trained machine learning models (random forests combined with dimensionality reduction methods) to discriminate between different types of 3D reach-to-grasp maneuvers based on muscle activity in the arm and forearm [12]. Their results showed that it is possible to accurately classify different grasping techniques by analyzing myoelectric patterns, highlighting the potential contained in EMG signals from the upper limb for characterizing functional movements. Approaches like that of Liarokapis et al. are promising for human-machine interfaces and emphasize the need to better understand the activation patterns underlying complex gestures.

Despite these specific advances, it is important that a reference muscle activation pattern for reach and grasp, applicable in clinical research settings, has not yet been standardized. The lack of a normative "map" negatively affects the interpretation of new findings in the EMG field outside the specific context of each study. Having a typical patter would offer multiple benefits: on one hand, it would allow for contrasting a patient's muscle activity with that of healthy subjects, identifying which muscles exhibit atypical activations; on the other hand, it would ease the follow-up process for neurological and musculoskeletal patients, by being able to compare their pre- and post-rehabilitation EMG recordings against an expected standard. In the neurorehabilitation field, the availability of normative curves for the reach and grasp gesture could serve to objectify recovery (for example, by quantifying how closely the pattern of a hemiparetic patient approaches the normal pattern after a certain period of therapy) and to characterize altered patterns typical of different pathologies (for instance, activating inefficient compensatory muscles after a spinal cord injury). Several authors have pointed out the potential clinical value of delving deeper into upper-limb movements: a better understanding of the muscular roles in these types of gestures would help improve rehabilitation protocols, optimizing the design of active prothesis and refining biomechanical models of the hand's function [2]. In conclusion, there is a clear need to systematize EMG analysis of the reach-and-grasp gesture by standardizing methods and collecting data from healthy subjects to establish a reference pattern useful for clinical practice and research.

1.6 Objectives

The main objective of this thesis is to quantitatively characterize the typical muscular activation pattern during a reach and grasp gesture on healthy subjects, by obtaining normalized average curves that describe the activity of each muscle throughout the complete reach and grasp cycle. To this end, EMG processing techniques similar to the ones used during gait analysis will be employed: a temporal normalization of the signals will be performed (expressing time as a percentage of the reach and grasp cycle, from the start of the reaching movement to the completion of the grasp), along with amplitude normalization (in this studies' case, relative to the maximum contraction of each muscle during the gesture). Subsequently, the mean curves and their corresponding variability bands (standard deviation) will be calculated from multiple repetitions of the gesture. These normalized curves will represent the typical behavior of the main muscles involved during the reach and grasp sequence in normal conditions.

Additionally, as a secondary objective, this thesis will explore the consistency of said patterns both across successive repetitions by the same person and among different subjects. This will involve evaluating the similitude level both intra- and inter-subject of the obtained muscle activation curves, determining the extent to which this pattern is reproducible across different trials and common among various healthy individuals. Verifying the existence of a strong and consistent pattern is an indispensable previous step for future clinical applications: it will establish confidence that the differences observed between a patient and the normal pattern are truly due to pathological alterations and not to random variability. In perspective, the results of this thesis will set the foundations for the creation of a reference reach and grasp EMG database, which could be used in the future to monitor the evolution of neurological patients, objectively quantify their rehabilitations progress, and detect atypical muscle activation patterns that guide therapeutic interventions in a more personalized manner. Finally, this work also aims to identify which muscles provide the most representative information during the execution of the reach and grasp gesture. This information would allow for a reduction in the number of electrodes needed, thereby optimizing the instrumentation used in future studies or clinical applications without compromising the quality of the analysis.

In summary, this project aims to take a first step towards standardizing the electromyographic analysis of the reach and grasp gesture, emulating the success of gait analysis, with the goal of improving the functional assessment of the upper limb in clinical settings.

2 MATERIALS AND METHODS

2.1 Materials

2.1.1 Trigno Avanti from Delsys

The surface electromyography (sEMG) system employed on this thesis was the Delsys Trigno Avanti (Delsys Inc., Boston, MA), a widely recognized wireless system for its precision and versatility in clinical and investigation applications [13] (Figure 1). This system consist of surface sensors that allow the measurement of electrical activity of the muscles through high-sensitivity adhesive electrodes.

Figure 1 - Delsys Trigno Wireless EMG System [14]

System characteristics:

- Type of sensors: wireless surface sensors.
- Sampling frequency: 2148 Hz per channel.
- Frequency range: 20-450 Hz, optimized for myoelectric activity measuring.
- Resolution: 16 bits.
- Connectivity: Wireless data transfer to base station, allowing freedom of movement for the subject.
- Software: EMGworks, that facilitates both the acquisition and export process of the signals.

2.2 Methods

2.2.1 Participants

Twenty-seven participants (13 men and 14 women) between the ages of 18 and 25 (mean age: 20.86 years ± 2.03), all right-handed were recruited for this study. None of them presented any musculoskeletal pathologies nor a history affecting the mobility of their dominant upper limb. Before the study started, each participant was informed about the procedure and gave their verbal informed consent. Participants were allowed to do a few unrecorded trials to familiarize themselves with the task and to ensure that none of the sensors interfered with their natural movement pattern.

2.2.2 Experimental procedure

Subject preparation: each participant adopted a seated position, with their back straight against the backrest and feet flat on the floor to standardize posture. The dominant arm (right) was positioned with the elbow at 90º flexion and the forearm in a neutral position, resting on an adjustable-height stool in front of the subject (as seen in Figure 2).

Figure 2 – Subject's initial posture

Electrode placement: The subject's skin was prepared at the measuring sites following SENIAM guidelines [15] for surface electromyography, their skin was cleaned with isopropyl alcohol to reduce impedance and noise. Next, bipolar silver surface electrodes were placed on the muscle bellies of interest, aligned with the direction of the muscle fibers and with approximately 20 mm of separation between electrodes (see Figure 3). The eight upper limb muscles that were evaluated consist of: biceps brachii, triceps brachii, anterior deltoid, lateral (middle) deltoid, posterior deltoid, upper trapezius, palmaris longus and extensor carpi radialis (see Figure 4) [16]. The exact placement of each electrode followed the specific recommendations of SENIAM to ensure optimal activity capture and reproducibility between subjects.

Figure 3 – Anterior, lateral, and posterior view of electrode placement

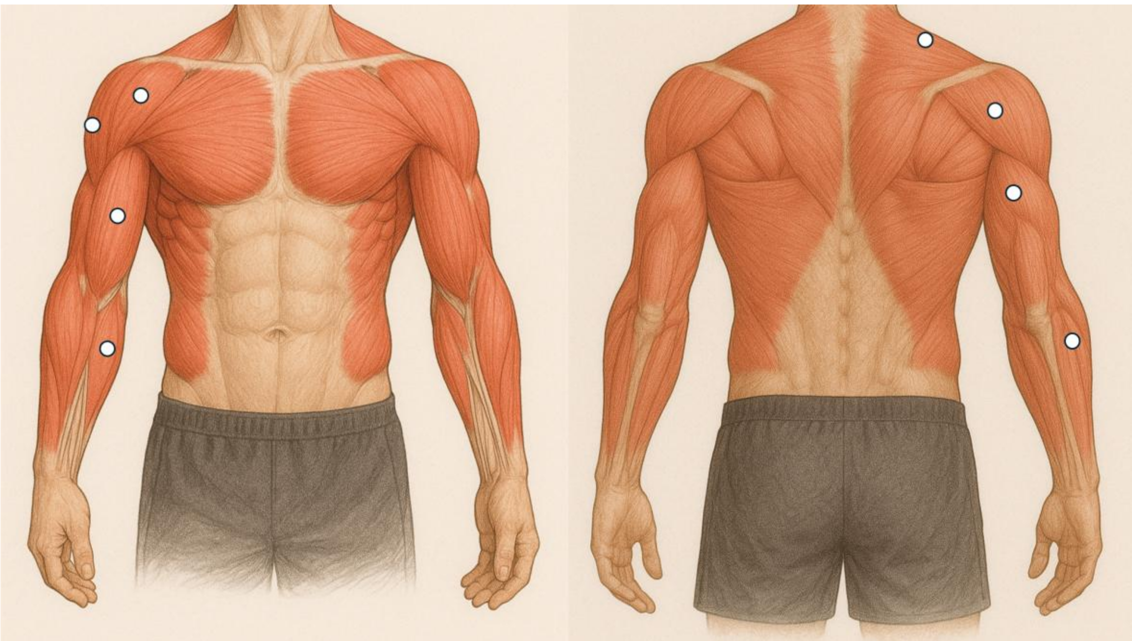

Figure 4 – Selected muscles named top to bottom, left to right: anterior deltoid, lateral (middle) deltoid, biceps brachii, palmaris longus, upper trapezius, posterior deltoid, triceps brachii and extensor carpi radialis

Motor task: After the subject's preparation, the participant was instructed to perform a functional "reach and grasp" movement, similar to a daily activity of reaching and grasping an object. The object employed in the study was an empty coffee cup placed at the right of the patient's resting hand. Each repetition of the task consisted on four phases: (1) reaching and grasping the glass, (2) transporting or moving the glass by lifting it up to their mouths, (3) transporting the glass from their mouths back to the stool in its initial position and (4) releasing the cup on the stool at its original location and returning their hand to the initial position at 90º. Participants performed this complete cycle 5 consecutive times each with a small rest between repetitions of a couple of seconds to allow the muscles to fully relax. This number was selected based on practical considerations, as it was deemed sufficient to generate a representative average while minimizing fatigue and preserving the natural execution of the movement. No external rhythm was imposed on the subjects to allow maximum comfortability and naturality to the movement, the only imposed constraint was to perform the movement in a fluid manner without stopping at any point during the repetition. Before the measurements, each subject was allowed to practice the movement to familiarize themselves with the task and ensure consistent execution. During the task, the participants were supervised to ensure all the above conditions were met avoiding compensation or other movements that could contaminate the EMG recordings. If any of the repetitions were deemed invalid due to execution errors or technical issues, an additional repetition was performed to ensure five valid trials were obtained per subject. In total, each session lasted approximately 30 minutes, including electrode placement, familiarization trials, and the recording of the functional task.

2.2.3 Data acquisition and preprocessing

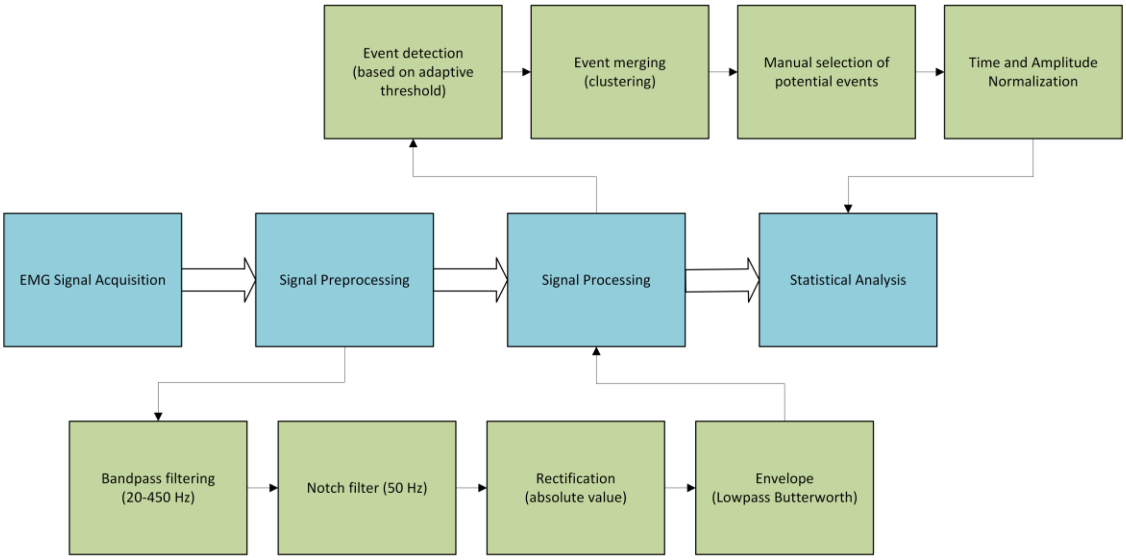

Figure 5 – Signal Processing Schematic

EMG recording system: The electromyographic activity of the eight muscles was recorded using the Delsys surface EMG system (Delsys Inc., Boston, MA) considered the gold standard for high-fidelity EMG acquisition. Eight simultaneous channels were employed (one for each muscle), acquired wirelessly and continuously during the execution of the functional task. Each signal was sampled at 2148 Hz with 16-bit resolution. The Delsys system is equipped with high-gain integrated amplifiers and common-mode rejection (>80 dB) to minimize interference. Raw EMG data (voltage over time) was stored for subsequent analysis in MATLAB (Mathworks 2024b, Natick, MA, USA).

Signal processing: The raw EMG signals were processed offline using a standardized pipeline analogous to that used in gait pattern analysis [17], with the aim of extracting a noise-free muscle activation envelope and making the signals comparable across repetitions and subjects. The processing included the following main steps (See Figure 5):

- Bandpass filtering (20-450 Hz): A digital 4th order Butterworth filter was applied to the 20-450 Hz band to get rid of low frequency components (motion artifacts and movement noise) and high frequency (electrical noise) outside the normal surface EMG range . The selected range encompasses the most relevant frequencies of the physiological EMG signal, preserving muscle activation information while attenuating noise. Filtering was performed using a zero-phase filter algorithm (using *filtfilt*) to prevent phase shifts in the processed signal.
- Notch filter (50 Hz): Additionally, a notch filter centered at 50 Hz (the mains electricity frequency) was implemented with an adjusted Q factor to suppress sinusoidal interference from the power line, accounting for the possible noise introduced by the European 50 Hz alternating current.
- Rectification: After the bandpass filter, each EMG signal was fully rectified, converting all values to positive (absolute value of the signal). Rectification transforms the bipolar signal into a unipolar one, reflecting the instantaneous magnitude of muscle activation regardless of polarity.
- Envelope calculation: The linear envelope of each rectified signal was obtained by applying a 4th-order low-pass Butterworth filter with a low cutoff frequency (~8–10 Hz). This filtering smooths the rapid fluctuations of the rectified signal, producing an envelope curve that represents the trend of muscle activation over time [18]. The result is a smooth curve that follows the changes in muscle contraction intensity throughout the movement (also known as linear envelope).

2.2.4 Movement segmentation

Segmentation of repetitions (delimiting each reach-grasp-release movement cycle) was performed semi-automatically using the preprocessed EMG signals (Figure 6). First, a custom adaptive threshold event detection algorithm was used on the envelope to identify the start and end points of each repetition. In essence, an activation threshold was calculated for each muscle channel based on the noise level and baseline activation of the signal. For instance, a threshold was defined as a high percentile (e.g., 90-95%) of the amplitude distribution of the envelope, adapted to each muscle. When a certain muscle's envelope surpassed this threshold, the start of a possible active phase (beginning of a reach) was marked when the signal exceeded the threshold, and the moment it fell back below the threshold it indicated the end of that phase. Since multiple muscles activate in a coordinated manner during a functional movement, the algorithm evaluates all channels for robustness: a new repetition was considered to begin when a set of muscles (such as the deltoids and biceps) simultaneously showed rising activity above their thresholds, and it ended when the activity dropped after the release phase. These automatic detections were then manually validated, ensuring that each identified segment effectively corresponded to a complete reach-grasp-release cycle. In case of erroneous detections or imprecise segmentations (for example, if an overly sensitive threshold marked sub-movements), the parameters were adjusted, or the markings were manually corrected. To improve the robustness of the segmentation process and avoid false detections caused by signal fluctuations, additional clustering step was implemented to group temporally close activation events. Once the initial candidate segments were identified, their central points were calculated and grouped using a hierarchical clustering algorithm (single-linkage method), with the estimated average duration of a typical repetition used as the distance threshold. This allowed for the merging of adjacent or overlapping events that likely belonged to the same functional cycle. For each resulting cluster, the earliest star and the latest end were selected as the new refined segment. In addition, segments whose duration did not meet a physiologically plausible threshold (100 ms) were discarded to eliminate spurious detections. This procedure ensured that each final segment corresponded to a complete and valid cycle of reaching, grasping, and releasing. Finally, a clear segmentation into five well-defined repetitions was obtained for each participant, allowing for the separate analysis of each movement cycle and the correct alignment of data for averaging comparisons.

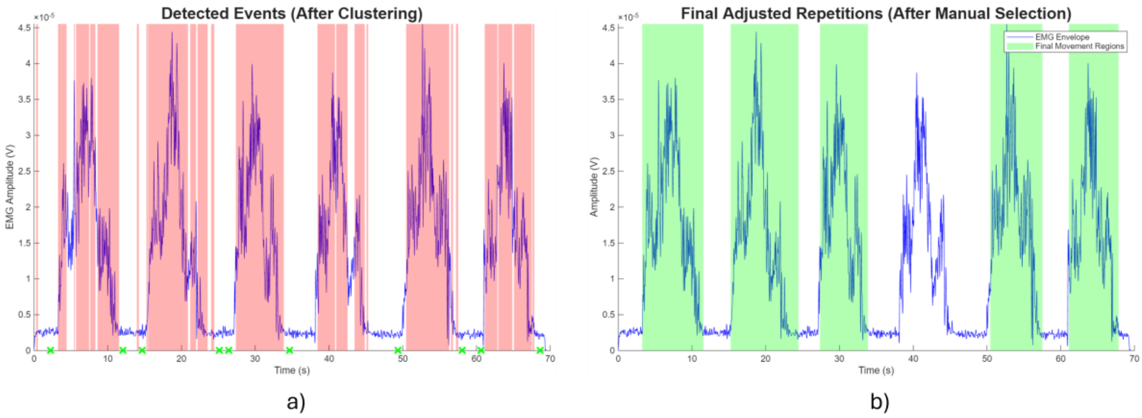

Figure 6 - Automatic event detection (a) and manual selection of automatically detected events (b)

2.2.5 Normalization

After the segmentation of the repetitions, each one of them was then normalized both in amplitude and in time (Figure 7):

- Temporal normalization: Given that the duration of each repetition may slightly vary between cycles and between subjects (since no fixed rhythm was imposed), a temporal normalization of the data for each repetition was performed. Specifically, the envelope of each repetition (from the start of the grasp phase till the end of the release phase) was resampled or interpolated to one thousand evenly spaced points. This way, each movement cycle is represented by the same number of signals, equivalent to 100% of the cycle duration. This procedure, inspired by gait analysis, allows time to be expressed on a common percentage scale, facilitating the comparison and averaging of cycles with different durations [18].
- Amplitude normalization: To be able to compare the EMG amplitudes between different subjects and muscles, an amplitude normalization of the envelope was performed. Each curve was normalized to a unitary scale (0 to 1) applying min-max normalization: each sample was adjusted by subtracting the minimum value of that channel and then dividing by the range (max – min). With this scaling, the value 1.0 corresponds to the maximum activation level observed in that muscle

(for that subject), and zero corresponds to the minimum level(rest/baseline) during the task. This normalization reduces the influence of individual differences in absolute amplitude due to factors such as muscle mass variations, electrode positioning, or contact quality, allowing the focus to be on the temporal activation pattern rather than absolute magnitude [19]. It is important to note that normalization was performed after obtaining the envelope and separately for each muscle and subject.

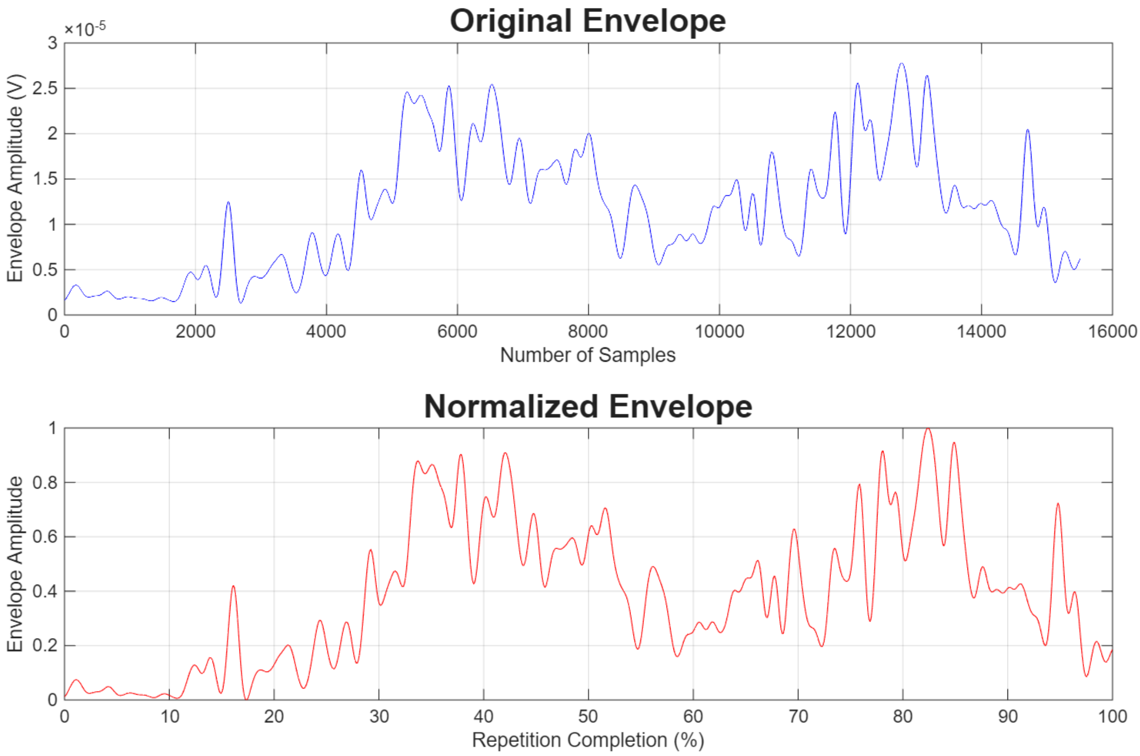

Figure 7 – Time and amplitude normalization visualization of a biceps brachii signal

2.2.6 Averaging of EMG repetitions for inter-subject analysis

After the temporal and amplitude normalization of each individual repetition, a representative average curve was generated for each muscle and subject. This step was conducted using a script which automated the process of loading, grouping, and combining the five repetitions for each subject.

For each muscle, the point-by point average of the normalized EMG signals was calculated, resulting in a single average activation curve per subject. This curve represents the typical activation pattern of that muscle during the reach and grasp task, limiting intra-subject variations inherent to repeated execution.

The procedure also computed the intra-subject standard deviation; this was later used for visualization purposes (Figure 8). Additionally, the combined results were saved into separate files which were then used for inter-subject analysis.

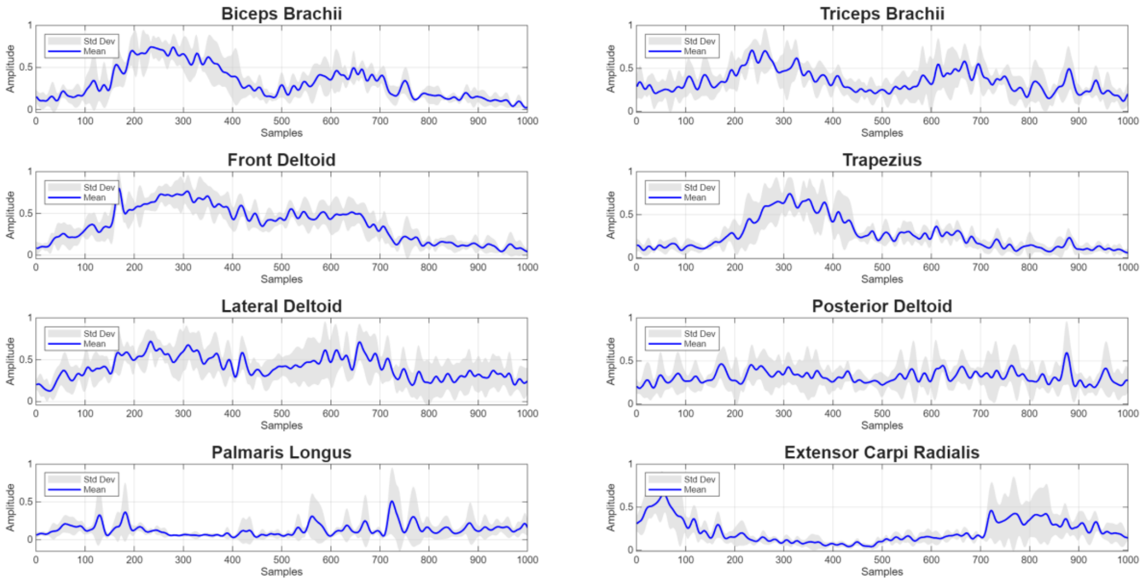

Figure 8 - Intra-subject average muscle activation curves for each muscle

2.2.7 Data analysis

The quantitative analysis was focused on assessing the intra-subject consistency of EMG patterns across repetitions as well as the similarities and inter-subject variability in these averaged patterns. For each subject and muscle, an average EMG curve was first calculated by point-to-point averaging of the five normalized repetitions for that subject (i.e., obtaining the mean muscle activation profile during the reach & grasp cycle for each subject). Using these individual average curves, the following analyses were performed:

- Intra-subject consistency: the reproducibility of each muscle's activation pattern within the same subject across five repetitions was assessed. To this end, the cross correlation was employed [10] between pairs of curves from the same person. In practice, for each subject and muscle, the mean correlation coefficient between each pair of repetitions was calculated (considering possible small temporal shifts to align peaks, although temporal normalization already approximately aligns them). These correlation coefficients (values between -1 and 1, where 1 indicates identical wave shapes) quantify the similarity of the EMG signal shapes across repeated attempts of the same movement. A high coefficient close to one indicates that the subject activated that muscle in a similar way across all repetitions (high consistency), while very low values indicate variations in coordination or relative intensity across repetitions. As a result, we obtained an average consistency measure for each muscle in each subject. This intra-subject metric allowed us to verify if subjects could reproduce a stable activation pattern, analogous to verifying the repeatability of gait patterns on consecutive trials.

- Inter-subject comparative: To examine differences or similarities between individuals, the correlation between the average curves of different subjects was analyzed. For each muscle, the average curve of each of the 27 participants was taken, and cross-correlation coefficients were calculated between each pair of subjects (i.e., comparing the shape of the curve of Subject A with Subject B, Subject A with Subject C, and so on). This process generated a 27 x 27 correlation matrix for each muscle, where each element (i, j) represents the degree of similarity in the EMG pattern between subject i and subject j. To enhance robustness, the correlation was calculated in shift mode: a small temporal offset (a few points on the normalized scale) was allowed when calculating the mean correlation. This adjustment ensured that small differences in the exact timing of activation do not unfairly penalize the comparison of global patterns. A high coefficient (>0.7) between two subjects indicates that they exhibit very similar activation patterns in that muscle during the gesture (e.g., they activate and relax the muscle at very similar phases of the movement). On the other hand, low correlations suggest differential patterns or distinct motor strategies between subjects. Beyond the complete correlation matrix, the mean and standard deviation of all inter-subject correlation coefficients (excluding the diagonal for perfect self-correlation) were calculated. These values provide a global summary of the similarities between participants for each muscle.

Technical note about cross-correlation implementation: In the present study, the cross-correlation technique was applied to assess the consistency of EMG patterns both intra-subject (between repetitions of the same individual) and inter-subject (between different individuals) based on their average patterns. For the intra-subject analysis, MATLAB's xcorr function was used to calculate the cross-correlation between each pair of trials for the same subject (considering each muscle separately), allowing a temporal lag of twenty-five samples to compensate for small misalignments in time between normalized curves. The correlation was computed in a normalized form (correlation coefficient), according to the following equation:

$$\hat{R}_{xy}(m) = \begin{cases} \displaystyle\sum_{n=0}^{N-m-1} x_{n+m} y_n^*, m \geq 0 \\ \hat{R}_{yx}(-m), m < 0 \end{cases}$$

Where x_n and y_n represent the EMG signals to be compared (already aligned in duration), m is the lag or temporal shift, and the asterisk denotes the complex conjugate (which is irrelevant in our case, as the signals are real). This calculation is repeated over a defined range of lags ($m \in [-25,25]$), generating a correlation function for each pair of signals. From each resulting correlation function, the mean value within the allowed lag range was extracted as a summary measure of the global similarity between two signals. Using all pairs of repetitions within the same subject and muscle, a correlation matrix of size $N \times N$ was constructed (where N is the number of repetitions), with each element representing the similarity between two trials. Finally, the mean and standard deviation of the coefficients (excluding the main diagonal) were calculated as estimators of the intra-subject stability of the muscle activation pattern. The inter-subject analysis was performed in a very similar way with the only change being that the correlation matrix was of size $M \times M$ (M being the number of subjects).

Overall, the analysis combined metrics of repeatability (within each individual) and comparability (between participants). This approach is equivalent to that used in gait analysis for comparing normalized muscle activation curves across different steps and subjects providing both validation of the stability of the collected EMG data and insights into the natural variability among individuals in performing the reach & grasp movement.

2.2.8 Results visualization

To better interpret and present the findings, graphical tools were used to highlight the average trends and variability of EMG data:

- Mean curves with variability: For each analyzed muscle, a graphical mean activation pattern was built along the movement cycle (0-100% of reach & grasp), obtained by averaging the five repetitions of each subject and the averaging again across all subjects. In these graphs, the main curve represents the mean activation (in normalized units) as a function of the percentage of the movement cycle, and a shaded area around the curve indicates ± std. dev of the signal at each point. This shading represents the variability across subjects for each phase of the movement: a small area (little deviation) indicates that most subjects shared a similar mean activation pattern at that moment, whereas a bigger area suggests greater dispersion in activation (for example, some activated earlier and some later). These figures clearly visualize the typical activation pattern of each muscle during the functional gesture, as well as how much variation exists around a typical pattern.
- Correlation matrices between subjects: As a numerical and visual aid to the inter-subject analysis, heatmaps of cross-correlation matrices between subjects were generated for each muscle. In these heatmaps, each axis represents the subjects (1 through 27) and each colored cell represents the correlation coefficient between the corresponding pair of subjects. A calibrated color scale was used (e.g., cool colors for low correlations to warm/intense colors for high correlations), making it easier to identify pairs or groups of subjects with high similarity in the EMG patterns. This type of visualization clearly reveals group tendencies – for instance, if most coefficients are elevated, it implies that all subjects share a common pattern; conversely, the presence of areas with consistently lower correlations would indicate subgroups of subjects with atypical patterns or significant differences in motor strategy.

3 RESULTS

3.1 Intra-subject cross correlation results

High levels of consistency were observed in the EMG signals of each subject across repetitions for each reach and grasp task. To quantify repeatability, the mean cross-correlation was calculated between each pair of repetitions from the same subject for each muscle (See Figure 9). To help compensate for small variations in temporal alignment of activation peaks between repetitions, a maximum lag of ±25 samples was allowed. This tolerance is especially useful to prevent slight phase shifts from negatively affecting the similarity value, focusing more on the overall shape of the EMG pattern rather than exact synchrony between repetitions. These results were then averaged across all participants, producing high and consistent values for all the muscles analyzed.

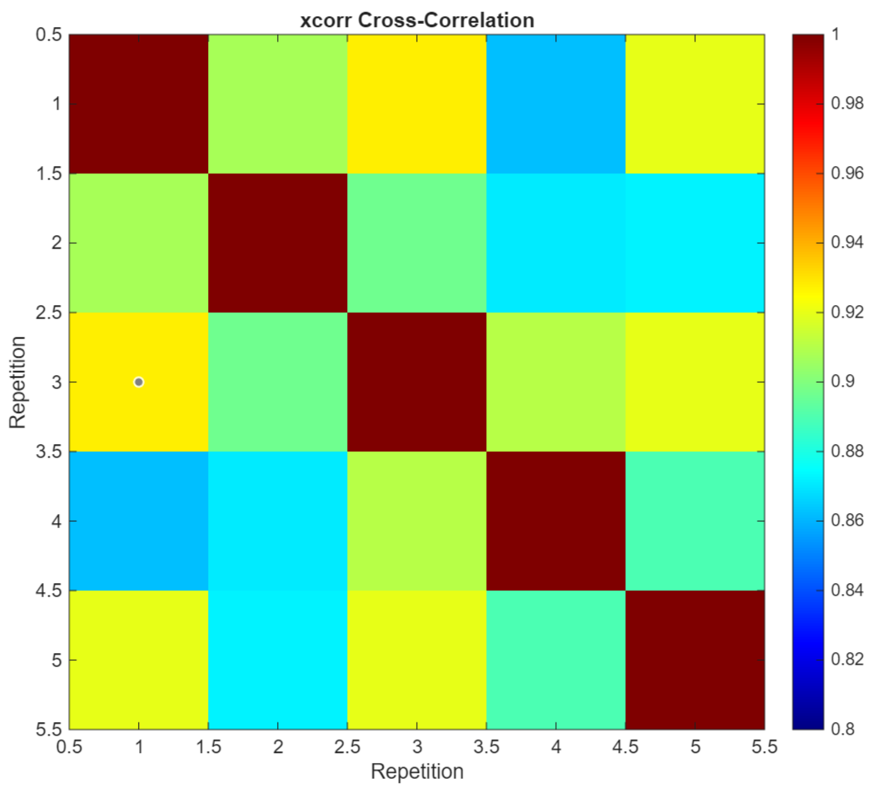

Figure 9 – Intra-subject cross-correlation heatmap

Table 1 presents the mean correlation coefficients ± standard deviation obtained between repetitions, calculated for each muscle based on the from the 27 subjects.

Muscles such as the front deltoid stand out with a mean value of 0.89 ± 0.09 and the trapezius with 0.87 ± 0.04, manifesting a remarkably high stability between repetitions. In contrast, the slightly lower values observed in the palmaris longus and extensor carpi radialis may reflect greater activation variability between consecutive cycles, although their values remain high (0.75 and 0.78 respectively).

Table 1 - Mean intra-subject cross-correlation (mean ± std) per muscle.

Muscle	Intra-subject correlation (Mean ± STD)
Biceps Brachii	0.85 ± 0.06
Triceps Brachii	0.81 ± 0.07
Front Deltoid	0.89 ± 0.09
Trapezius	0.87 ± 0.04
Lateral Deltoid	0.86 ± 0.03
Posterior Deltoid	0.81 ± 0.06
Palmaris Longus	0.75 ± 0.08
Extensor Carpi Radialis	0.78 ± 0.09

3.2 Inter-subject cross correlation results

At the inter-subject level, the muscle activation patterns also showed a high degree of similarity between different participants for each of the muscles analyzed. To quantify this similitude, the mean cross correlation was calculated between the average curves of each subject for the same muscle, allowing a maximum lag of ±25, as previously explained. This strategy allowed us to evaluate not only the exact coincidences between curves, but rather the global shape similarities in the temporal activation profiles.

Table 2 presents the average inter-subject cross correlation coefficients (excluding the main diagonal) and their standard deviation for each of the eight evaluated muscles. These results evidence a high general consistency among subjects, with correlation values that oscillate between 0.80 (Palmaris Longus) and 0.83 (Triceps Brachii). These values suggest the existence of a shared activation pattern between most participants during the functional task execution, especially in muscles such as triceps where correlations were particularly high and standard deviations low.

Table 2 – Average inter-subject correlation (mean ± STD) per muscle

Muscles	Inter-Subject Correlation (Mean ± STD)
Biceps Brachii	0.80 ± 0.05
Triceps Brachii	0.83 ± 0.04
Front Deltoid	0.81 ± 0.05
Trapezius	0.81 ± 0.05
Lateral Deltoid	0.80 ± 0.06
Posterior Deltoid	0.82 ± 0.04
Palmaris Longus	0.80 ± 0.06
Extensor Carpi Radialis	0.80 ± 0.05

In addition, these findings were then visualized using heatmaps and average activation curves for each muscle. Figure 10 shows the heatmaps of the inter-subject correlation matrices for each muscle, where each map illustrates the correlation coefficients between the normalized EMG signals of that muscle across all subject pairs. It can be observed that, for all muscles, the heatmaps display high correlation coefficients for virtually every pair of subjects. This quantitatively reinforces the similarity of muscle activation patterns across different individuals.

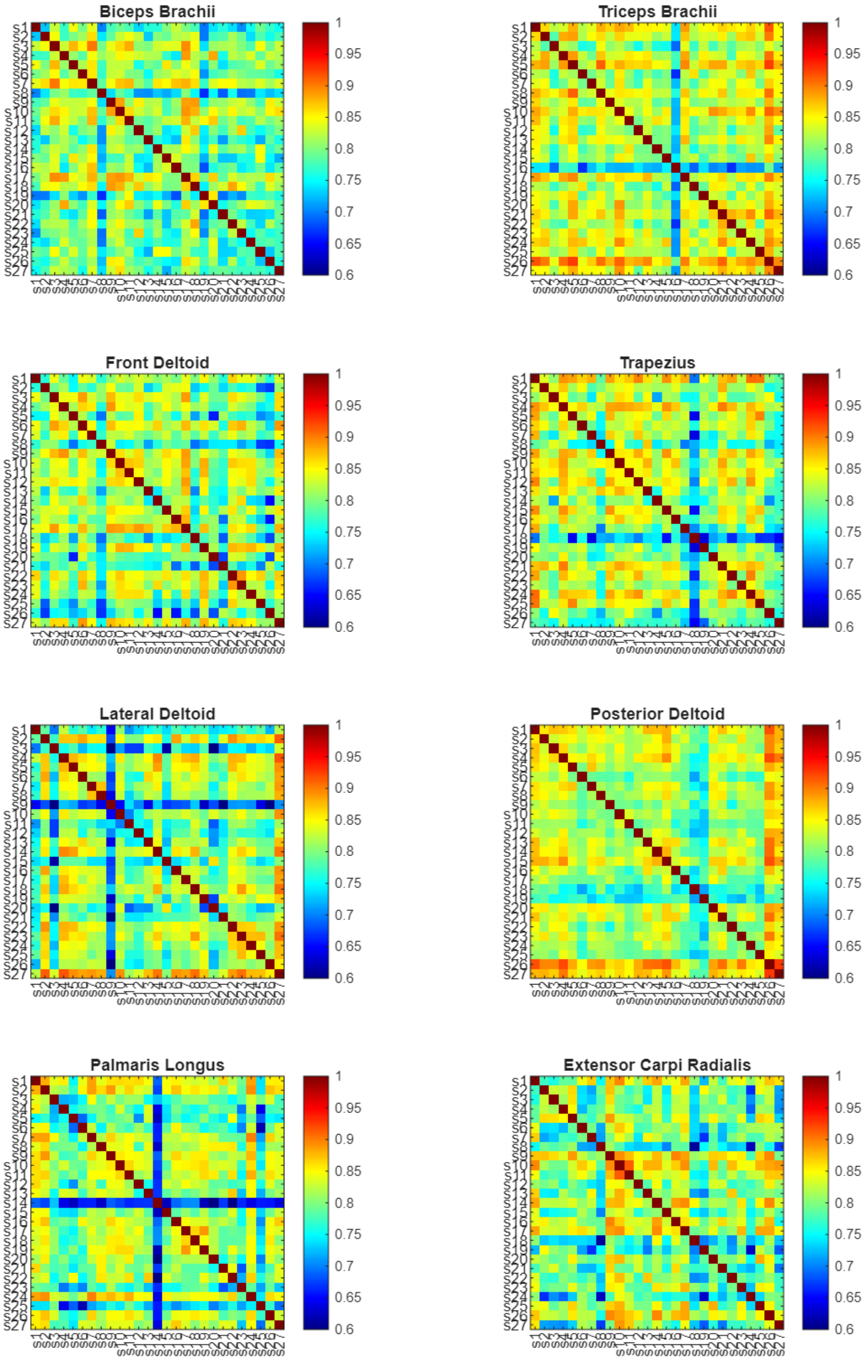

Figure 10 - Heatmaps of the inter-subject cross-correlation matrices for each muscle.

On the other hand, Figure 11 shows the normalized mean EMG activation curves for each muscle throughout the full movement cycle (from 0% to 100% of the reach and grasp task). To build these curves, each EMG signal was normalized to its maximum value, and the average activation of each muscle was calculated as a function of the movement cycle percentage for each subject. Then, the individual curves from all subjects were averaged to obtain a group-level temporal profile for each muscle. The resulting curves demonstrate that the temporal activation shape is similar across subjects for each muscle, with small amplitude variations.

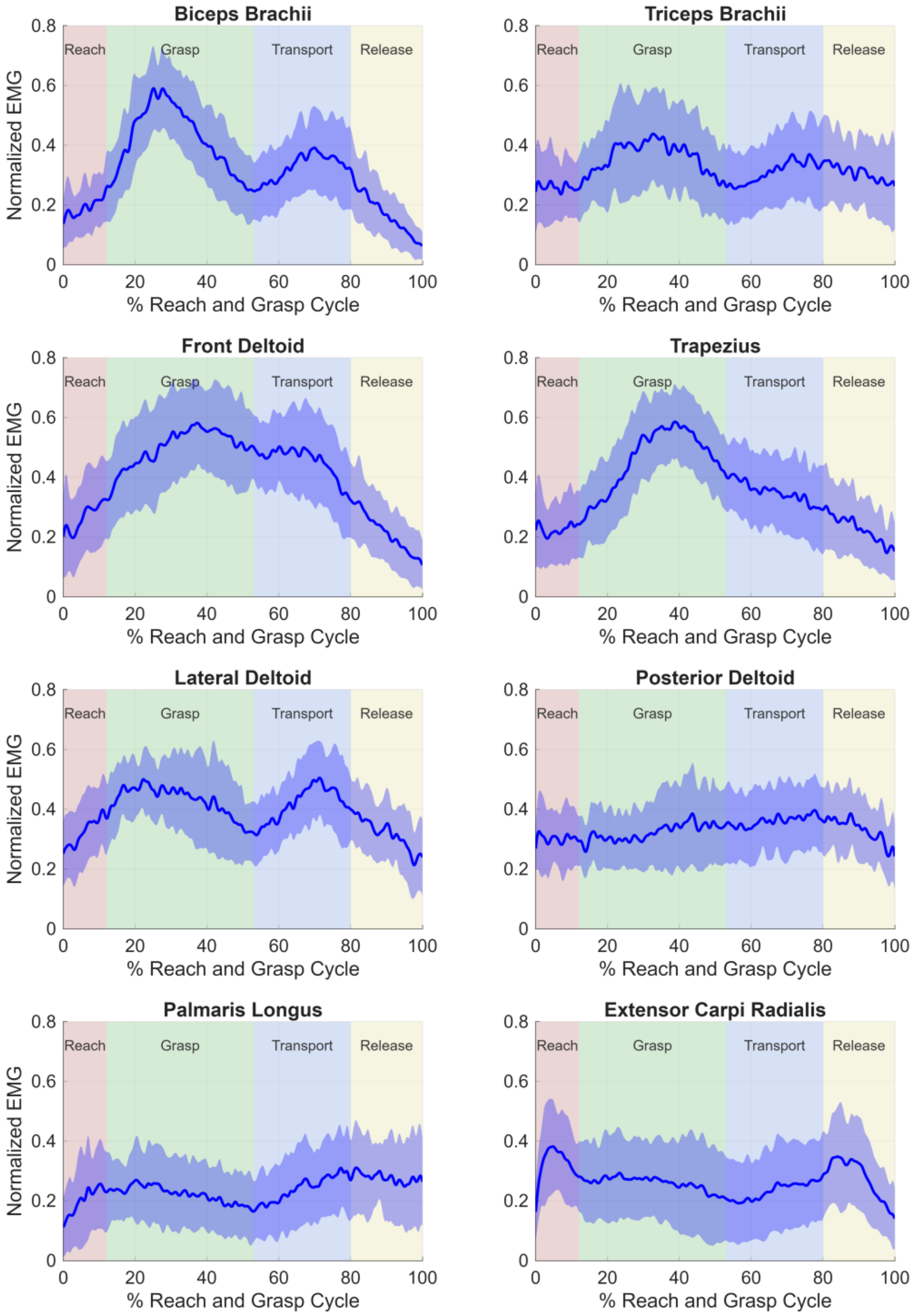

Figure 11 - Normalized means EMG and shaded STD activation curves by muscle throughout the reach and grasp movement cycle.

4 DISCUSSION AND CONCLUSIONS

The obtained results evidence a notable consistency in the muscle activation patterns measured through EMG during the performance of the functional reach and grasp task in healthy subjects. Intra-subject wise, high levels of reproducibility were observed: the cross correlation between different repetitions for the same subject yielded high mean values for all the muscles analyzed (Table 1). Proximal muscles such as the front deltoid (0.89 ± 0.09) and trapezius (0.87 ± 0.04) stood out for their high stability across repetitions. In contrast, the forearm muscles showed slightly lower values – for example, palmaris longus (~0.75) and extensor carpi radialis (~0.78) – suggesting greater variability in their activation across consecutive cycles. Nevertheless, even these lower coefficients fall within a relatively high range, indicating that each subject tends to repeat the muscle activation sequence quite consistently during each execution of the task.

Similarly, at the inter-subject level, a clear temporal muscle activation pattern was found to be shared among different individuals. The average cross-correlation between the activation profiles of each subject pair (calculated for each muscle) ranged approximately between 0.80 and 0.83 (Table 2). These high values, along with reduced standard deviations, highlight the presence of a common muscle activation pattern during the execution of the reach and grasp task in the healthy population we evaluated. Particularly notable is the case of the triceps, which showed the highest inter-subject correlation accompanied by a low variability, suggesting a fundamental and highly consistent role of this muscle in the functional kinematics of the task. On the other hand, the palmaris longus exhibited the lowest inter-subject correlation, indicating that the activation of certain distal muscles may vary more from one individual to another – possibly reflecting different strategies for stabilizing the wrist and hand during grasping. Despite these small differences, the overall high degree of similarity suggests that most participants relied on a homogeneous neuromuscular strategy to complete the movement.

The visual analyses reinforce and complement these quantitative findings. Figure 10 shows heatmaps of the inter-subject correlation matrices for each muscle, in which high coefficients can be observed for virtually every pair of subjects considered. In other words, all individuals present a high activation pattern similitude among themselves, which reinforces in a visual and quantitative way the inter-subject consistency we just described. Figure 11 presents the normalized mean EMG activation curves for each muscle throughout the full movement cycle (0-100%). These group average curves reveal that the temporal shape of activation is remarkably similar across all subjects, with relatively small amplitude variations. A well-defined muscle activation pattern can be observed: for example, the shoulder muscles and triceps are predominantly activated during the initial phases of arm extension, while the biceps brachii reaches its peak activation during elbow flexion as the cup is brought to the mouth; subsequently, in the return phase as the object is placed back on the table, triceps activity increases again (along with posterior deltoid) to control the extension and positioning of the arm. Finally, the forearm muscles (palmaris longus and extensor carpi radialis) are mainly activated both at the start and end of the movement when the cup is being picked up and deposited back on the table, although they are active throughout the whole movement as they must apply a certain continuous force for the cup to remain in the subject's hand. This sequential and coordinated pattern aligns with the expected biomechanical function of each muscle group during the subphases of the movement, and its reproducibility across subjects underscores the presence of common motor synergies in complex functional tasks.

From both a scientific and clinical standpoint, achieving a quantitative representation of a reach and grasp movement using only surface EMG is of great significance. Traditionally, the detailed characterization of movements has relied heavily on marker-based motion capture systems, such as Vicon, which provide precise kinematic measurements but require bulky, expensive equipment and controlled laboratory environments. In contrast, our approach demonstrates that it is possible to extract rich information about muscle coordination during movement using only EMG signals, without the need to rely on such complex systems. This represents a significant advantage in terms of simplicity and portability: EMG instrumentation is relatively easy to deploy in real clinical settings, allowing for patient assessment in a consultation room or even in a home environment. In this way, it eases the access to quantitative functional assessment tools by eliminating the need for specialized and costly technologies, thereby enabling broader adoption of these methods in daily clinical practice. Likewise, EMG offers a perspective that complements pure kinematics: it provides direct information about underlying neuromuscular activation, capturing aspects of motor control – such as recruitment sequence or muscle coactivation – that may go unnoticed when analyzing only external movements. This ability to directly quantify muscle function makes the proposed method a potentially very valuable tool for monitoring rehabilitation progress and neuromuscular disorders.

The analysis model developed opens the door to various practical applications. One of them is the objective comparison of muscle activation patterns before and after a surgical procedure or therapeutic intervention. For instance, in a patient who has undergone upper limb orthopedic surgery, the methodology would allow for the quantification of changes in their movement pattern during the post-surgical reach and grasp comparted to the pre-surgical state, identifying improvements in the muscle coordination or compensatory strategies that may have emerged after the intervention. Similarly, this approach is applicable to the analysis of individuals with motor disabilities or neuromuscular disorders. In patients who have suffered a cerebrovascular accident (stroke), incomplete spinal cord injury or other conditions that affect upper limb control, EMG characterization would provide objective metrics showing how a person's activation pattern deviates from that of a healthy individual. This can help quantify functional deficits, guide personalized rehabilitation programs (focusing on the muscles or movement phases that are the most altered), and monitor recovery over time. Even in progressive neuromuscular diseases (i.e., Multiple Sclerosis, Parkinson's Disease), this type of analysis could be used to detect subtle changes in muscular control during functional tasks, serving as an early indicator of deterioration or of treatment efficacy. In sum, having a normalized reference pattern of reach and grasp – obtained with an accessible technique such as surface EMG – offers a framework for comparing and assessing motor performance across a wide range of clinical scenarios. In addition to the consistency analyses, one of the secondary objectives of this study was to explore which muscles provide the most representative information for the entire gesture. This question is particularly relevant for reducing the required instrumentation, enabling the development of more streamlined and comfortable protocols for both researchers and clinical practice. Based on results, the deltoid muscles (anterior and middle), triceps, and biceps brachii showed consistent activation profiles with strong discriminative power across movement phases, suggesting they could serve as a minimal subset of electrodes for monitoring this type of functional task. The ability to limit instrumentation without compromising analytical capability would facilitate the integration of this methodology into clinical setting, where speed and ease of setup are key factors.

Despite the positive findings, it is important to acknowledge the study's limitations and propose future lines of research. First, the sample analyzed consisted solely of healthy young adults; therefore, the results obtained represent an idealized pattern of motor control that may differ in older populations or those with pathologies. It would be advisable to extend this type of analysis to such groups to examine how activation sequences and correlation values change (for example, it is expected that neurological patients would show lower inter-subject correlations with the typical pattern, due to compensatory strategies or spasticity). Secondly, the task studied is very specific and controlled (reaching for a glass/cup, bringing it to the mouth, and returning it to its original position). While it is representative of a daily activity, real-world conditions may vary (different movement speeds, object sizes and weights, individual grasping strategies, etc.), which could influence muscle activation. Future studies could evaluate task variations (e.g., different objects or contexts) to determine the robustness of the identified pattern and its adaptability. Another inherent limitation is the use of surface EMG, a technique that is susceptible to noise and interference. Factors such as electrode displacement, background noise and the superposition of adjacent muscle's signals (crosstalk) could affect the quality of the data. In this study, filtering and normalization procedures were applied to mitigate these effects, but they remain aspects that could be further optimized. Likewise, our analysis focused on the temporal shape of the signals (through normalization and correlation), prioritizing pattern similarity over differences in the absolute magnitude of activation. This implies that we did not explore in detail variations in EMG signal intensity between subjects or across repetitions, which could be relevant for assessing, for example, muscle effort or fatigue. To address this, future research could incorporate amplitude-based metrics or relative activation measures, as well as more precise calibrations to compare activation levels between individuals. Likewise, it would be valuable to complement the current approach with analyses of muscle synergies and dimensionality reduction techniques (e.g., principal component analysis or non-negative matrix factorization) that allow motor control to be described in terms of coordinated activation modules. This would provide a deeper understanding of the neuromuscular organization of movement and could reveal subtle differences not captured by linear correlation analyses. Finally, it is recommended to explore the integration of EMG data with kinematic information or other modalities (such as inertial sensors) to achieve a more comprehensive functional assessment. For example, combining EMG with motion capture could directly link specific phases of the gesture (e.g., reaching speed, grip stability) with the corresponding muscle activity, enriching clinical interpretation.

In conclusion, this study successfully characterized the functional movement of reaching and grasping an object using only surface EMG signals, demonstrating both high intra- subject repeatability and remarkable consistency in the activation pattern among different healthy subjects. The identification of this common muscular activation profile for the reach and grasp task highlights the ability of EMG to objectively and reliably describe the underlying neuromuscular coordination involved in daily life activities. The findings not only confirm that it is feasible to replace (or complement) kinematics-based analyses with EMG measurements for this type of task but also lay the groundwork for highly relevant clinical applications. Looking ahead, the methodology presented could evolve into an accessible and practical tool for functional assessment in rehabilitation, enabling the monitoring of motor recovery and the personalization of treatments based on the direct reading of the patient's muscle activity.

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NOTE: ChatGPT tool was used to create Figure 4, to assist in correcting code errors when they could not be found on other websites, as well as for refining the writing and translation.